

Review

$$1 \sin(A-B) = \sin A \cos B - \sin B \cos A$$

$$\therefore \sin(\theta - 4\theta) = -\sin 3\theta$$

$$2 \quad \therefore \frac{dr}{dt} = 0.005 \text{ ms}^{-1} = v$$

$$\text{light intensity} = \left(\frac{1}{r}\right)^2$$

$$\text{want } \frac{dI}{dt} = \frac{dI}{dr} \cdot \frac{dr}{dt}$$

$$I = 100 r^{-2}$$

$$\frac{dI}{dr} = -200 r^{-3}$$

$$\therefore \frac{dI}{dt} = -200 r^{-3} \cdot 0.005$$

$$= \underline{\underline{-0.008}}$$

$$3 \quad 24, -12, 6, -3$$

$$r = -\frac{1}{2}$$

$$\therefore u_n = 24 \cdot \left(-\frac{1}{2}\right)^{n-1}$$

$$S_n = \frac{24(1 - (-\frac{1}{2})^n)}{\frac{3}{2}} = 16(1 - (-\frac{1}{2})^n)$$

$$a) \quad 24 \cdot \left(-\frac{1}{2}\right)^6 = \frac{3}{8}$$

$$S_\infty = \frac{24}{1 + \frac{1}{2}}$$

$$= 16$$

$$\therefore \text{of } S_\infty = 15.84$$

$$S_n \geq 15.84$$

$$16(1 - (-\frac{1}{2})^n) \geq 15.84$$

$$(0.5)^n \leq 0.01$$

$$n \log 0.5 \leq \log 0.01$$

$$\therefore n \geq \frac{\log 0.5}{\log 0.01}$$

$$\geq 6.6$$

$$\therefore \underline{\underline{n=7}}$$

$$4 \int (x^2 + 2x)^{-1} dx$$

$$\frac{1}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2}$$

$$A(x+2) + Bx = 1$$

$$A = B$$

$$2A = 1$$

$$\therefore A = \frac{1}{2} \therefore B = -\frac{1}{2}$$

$$\therefore \frac{1}{x(x+2)} = \frac{1}{2x} - \frac{1}{2x+2}$$

$$\int (2x)^{-1} - (2x+2)^{-1} dx$$

$$= \frac{1}{2} \ln x - \frac{1}{2} \ln(x+2)$$

$$= \frac{1}{2} \ln \left| \frac{x}{x+2} \right| + c \quad (b)$$

$$5 \quad t = \sqrt{\frac{x}{5}}$$

$$y = 10 \sqrt{\frac{x}{5}}$$

$$y^2 = 20x \quad \subseteq$$

Consolidation

$$1 \quad x^2 + 1 = 2x, 1$$

$$x(x-2) = 0$$

Intervals at $x=0$ & $x=2$

$$\text{Wants: } \int_0^2 x^2 + 1 dx = \int 2x + 1 dx$$

$$= \int_0^2 x^2 - 2x dx$$

$$\left| \left[\frac{1}{3}x^3 - x^2 \right]_0^2 \right|$$

$$= \left| \frac{8}{3} - 4 \right|$$

$$= \left| -\frac{4}{3} \right|$$

$$= \frac{4}{3}$$

$$2 \quad x^2 + x + 1 = -x^2 + 2x + 5$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

Intervals @ $x=2$ and $x=-1$

$$\int_{-1}^2 x^2 - x - 2 dx = \left[\frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x \right]_{-1}^2$$

$$= \left(\frac{8}{3} - 2 - 4 + \frac{1}{3} - \frac{1}{2} + 2 \right) = \left(-\frac{3}{2} \right) = \frac{3}{2}$$

$$3 \quad y = \sqrt{4-x}$$

$$A = (0, 3)$$

$$y^2 = 4 - 4x$$

$$y^2 - 4 = -4x$$

$$4 - y^2 = 4x$$

$$x = \frac{4-y^2}{4}$$

$$\begin{aligned} \int x \, dy &= \frac{1}{4} \int_0^3 4 - y^2 \, dy = \frac{1}{4} \left[4y - \frac{1}{3} y^3 \right]_0^3 \\ &= \frac{1}{4} (27 - 4) \\ &= \frac{9}{2} \end{aligned}$$

$$4y' = 3x^2 + 4x - 4$$

$$\text{At } k, y' = 0$$

$$3x^2 + 4x - 4 = 0$$

$$(3x - 2)(x + 2) = 0$$

$$\text{At } x = -2, y = 4$$

$$\therefore k = 9 \quad (-2, 4) \text{ is where it touches.}$$

$$\text{i) } x^3 + 2x^2 - 4x + 1 = 9$$

$$x^3 + 2x^2 - 4x - 8 = 0$$

$$\text{At } x = 2, 1 = 0$$

$$\text{So, intersects @ } x = 2$$

$$\text{Other coordinate: } (2, 9)$$

$$\text{ii) } A = \left[\frac{1}{2} x^4 + \frac{2}{3} x^3 - 2x^2 + x \right]_{-2}^2$$

$$= \frac{16}{2} + \frac{32}{3}$$

$$= \frac{44}{3}$$

$$5) \text{ At } x = 0, y = t - t^2 = 0$$

$$t(1-t) = 0$$

$$t = 0 \text{ or } t = 1$$

$$6) \int y \, dt = \frac{1}{2} t^2 - \frac{1}{3} t^3$$

$$\int_0^1 y \, dt = \frac{1}{2} - \frac{1}{3} - 0$$

$$= \frac{1}{6}$$

$$6 \quad 1 - \cos \theta = 0$$

$$\cos \theta = 1$$

$$\theta = 0 \text{ or } 2\pi$$

$$\int_0^{2\pi} 1 - \cos \theta \, d\theta$$

$$= [\theta - \sin \theta]_0^{2\pi}$$

$$= 2\pi - 0$$

$$= 2\pi$$

$$7 \quad t = 0 \text{ or } t = 1 \text{ when } y = 0$$

$$\text{At } t = 0, x = 0$$

$$\text{At } t = 1, x = 4$$

$$x \text{ is also 4 at } t = -1$$

$$\int_{-1}^1 x(t) \cdot t^3 \cdot \sqrt{2} \, dt$$

$$\int_{-1}^1 16t^2 - 16t^4 \, dt$$

$$16 \left[\frac{1}{3} t^3 - \frac{1}{5} t^5 \right]_{-1}^1$$

$$\left[\frac{1}{3} - \frac{1}{5} + \frac{1}{3} - \frac{1}{5} \right]$$

$$16 \left[\frac{2}{3} - \frac{2}{5} \right]$$

$$= \frac{64}{15}$$

Exam Question

$$\begin{aligned} | \quad A \log x &= \ln 5 & A \log x &= \ln 12 \\ \ln 5 &= \ln(t^2 - 4) & \ln 12 &= \ln t^2 - 4 \\ 5 &= t^2 - 4 & t^2 - 4 &= 12 \\ t^2 &= 9 & t^2 - 16 &= 6 \\ t &= \pm 3, \text{ but } t > 2, \therefore t = 3 & (t-4)(t+4) &= 0 \end{aligned}$$

$$\begin{aligned} \text{Area of } R &: \int y \frac{dy}{dx} dx & t &= \pm 4 \\ & & t > 2, \therefore t &= 4 \\ &= \int_2^4 \frac{4}{t^2} \cdot \frac{2t}{t^2 - 4} dt & & \\ &= \int_3^4 \frac{8t}{t(t^2 - 4)} dt & & \\ &= \int_3^4 \frac{8}{t(t^2 - 4)} dt & & \\ & \quad a = -3, \quad b = 3 \end{aligned}$$

$$\begin{aligned} 6) & \int_3^4 \frac{1}{t(t+2)(t-2)} dt \\ \frac{A}{t} + \frac{B}{t+2} + \frac{C}{t-2} &= \frac{8}{t(t+2)(t-2)} \end{aligned}$$

$$A(t+2)(t-2) + Bt(t-2) + Ct(t+2) = 8$$

$$\begin{aligned} -4A &= 1 & 8B &= 1 & 8C &= 1 \\ A &= -\frac{1}{4} & B &= \frac{1}{8} & C &= \frac{1}{8} \end{aligned}$$

$$\begin{aligned} & \int_3^4 \frac{1}{t+2} + \frac{1}{t-2} - \frac{2}{t} dt \\ &= \left[\ln|t+2| + \ln|t-2| - 2\ln|t| \right]_3^4 \\ &= \ln 6 + \ln 2 - 4\ln 2 - \ln 5 - \ln 1 + 2\ln 3 \\ &= \ln 6 - 3\ln 2 - \ln 5 + 2\ln 3 \\ &= \ln 6 - \ln 8 - \ln 5 + \ln 9 \\ &= \ln \left(\frac{6 \times 9}{8 \times 5} \right) \\ &= \ln \left(\frac{54}{40} \right) = \ln \left(\frac{27}{20} \right) \\ k &= \frac{27}{20} \end{aligned}$$

$$2 \quad x = \frac{1}{4} \sin t$$

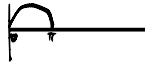
$$y = t \cos t$$

$$@ \quad k, \quad y = 0$$

$$t \cos k = 0$$

$$k = 0 \text{ or } \cos k = 0$$

$$k = \frac{\pi}{2}$$



a)

$$b) \quad \frac{dy}{dt} = t \times -\sin t + 1 \times \cos t$$

$$= \cos t - t \sin t$$

$$At \text{ max, } \cos t - t \sin t = 0$$

$$\text{When } t \text{ is small, } \cos t = 1 - \frac{t^2}{2}$$

$$\sin t = t$$

$$\cos t - t \sin t \approx 1 - \frac{t^2}{2} - t^2$$

$$1 - \frac{t^2}{2} - t^2 = 0$$

$$1 - \frac{3}{2}t^2 = 0$$

$$\frac{3}{2}t^2 = 1$$

$$t^2 = \frac{2}{3}$$

$$t = \sqrt{\frac{2}{3}}$$

$$c) \quad x = \frac{1}{4} \sin t$$

Using small angle approx:

$$x \approx \frac{1}{4} t$$

$$At \text{ P, } x \approx \frac{1}{4} \times \sqrt{\frac{2}{3}}$$

$$\approx \sqrt{\frac{2}{3}} \times \frac{1}{4}$$

$$\approx \sqrt{\frac{1}{24}}$$

$$\approx \frac{1}{2\sqrt{6}}$$

$$d) \quad i) \text{ finite region } R = \int_0^{\frac{\pi}{2}} y \frac{dx}{dt} dt$$

$$= \int_0^{\frac{\pi}{2}} t \cos t \times \frac{1}{4} \cos t dt$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{2}} t \cos^2 t dt \quad \left[\text{Note: } \cos^2 t = 2 \cos^2 t - 1 \right]$$

$$\frac{1}{4} \int_0^{\frac{\pi}{2}} t \left(\frac{\cos^2 t + 1}{2} \right) dt$$

$$= \frac{1}{8} \int_0^{\frac{\pi}{2}} t (1 + \cos^2 t) dt$$

$$a = \frac{\pi}{2}$$

$$b = \frac{1}{2}$$

$$(i) \frac{1}{8} \int_0^{\frac{\pi}{2}} t(1+\cos 2t) dt$$

$$\frac{1}{8} \left[\frac{1}{2} t^2 + \int t \cos 2t dt \right]_0^{\frac{\pi}{2}}$$

$$\dots + \frac{1}{2} \sin 2t \times t - \frac{1}{2} \int \sin 2t dt$$

$$+ \frac{1}{2} \left(\frac{1}{2} \cos 2t \right)$$

$$= \frac{1}{8} \left[\frac{1}{2} t^2 + \frac{1}{2} t \sin 2t + \frac{1}{4} \cos 2t \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{16} \left[t^2 + t \sin 2t + 2 \cos 2t \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{16} \times \left(\frac{\pi^2}{4} + 0 - 2 - -2 \right)$$

$$= \frac{\pi^2}{64} - \frac{1}{4}$$